Report Nº 2
Team Out of Lane
Date 19.01.2025

1 Introduction

This report details the activities carried out to date in the second sprint, with emphasis on linking input data to a rough output. These activities were focused on:

1. Adding hardware and the logic necessary for its implementation, such as ultrasonic sensors for emergency braking.
2. Start implementing the lane detection module.
3. Review and adapt our architecture and planning to the needs of the project and the team members.

2 Planned activities

Category	*Activity*	*Responsible*
Hardware improvements	Mechanical calibration	Alexander P.
	Research GPU addition	Alexander P. and Evelyn C.
	Research hardware additions (ultrasonic sensors and headlight)	Alexander P., Carolina N. and Evelyn C.
Computer vision package (RPi)	Implement lane detection	Alexander P. and Pablo F.
	Implement traffic sign detection	Evelyn C., Pablo F. and Victor P.
Project Documentation	Architecture revision	Pablo F.
	Architecture diagram	Evelyn C. and Pablo F.
	Prepare traffic sign dataset	Alexander P., Evelyn C. and Victor P.
	Document the 2nd report	Carolina N. and Pablo F.

3 Status of planned activities

Activity	*% completed*	*Observations*
Mechanical calibration	75%	We have fixed some hardware problems but there is still a slight deviation in the steering.
Research GPU addition	75%	We are still trying to acquire a GPU that fits our use cases and available resources. So far, our main options are: ▪ Jetson Orin Nano ▪ RPi AI Kit
Research hardware additions (ultrasonic sensors and headlight)	75%	We have acquired and installed two ultrasonic sensors and implemented their logic. We are still working on the headlights.
Implement lane detection	25%	We have developed a lane detection algorithm using OpenCV, but it struggles with certain points of the track (i.e.: crosswalk).
Implement traffic sign detection	10%	We have tested the integration of YOLO and the RPi.
Architecture revision	100%	We have reviewed and adapted our architecture to the needs of the project
Architecture diagram	100%	We have adapted our draw-io diagrams.
Prepare traffic sign dataset	50%	We have printed the signals and started preparing a dataset with them.
Document the 2nd report	100%	The report itself.

4 General status of project

The team has successfully captured input data from the ultrasonic sensors and gotten the vehicle to answer it. Major tasks focused on hardware additions and the embedded system have been completed. On the other hand, we have experienced a slight delay in tasks related to image processing, though a prototype of a lane detector has already been implemented.

The car be moved manually using the web application or on its own, being able to detect when something is in its path (thanks to the ultrasonic sensors) and stop to avoid hitting it, or even, if possible, reduce speed to continue moving forward with caution.

5 Upcoming activities

To be continued from the 2nd sprint

- **Mechanical calibration**
- **Research GPU additions**
- **Research hardware additions (headlight)**
- **Implement lane detection**
- **Implement traffic sign detection**
- **Prepare traffic sight dataset**

New activities

- **Define other necessary sensors, define use-case, integration (IMU, distance), preprocessing, noise cancelling**
- **Define use-cases and test given servers information (localization on map, cars interaction, GPS interaction, interaction with objects)**
- **Intersection detection**
- **Implement traffic light detection & classification**
- **Architecture revision**
- **Architecture diagram**
- **Define decision making (priorities of actions and state flow)**
- **Document the 3rd report**